

Optimization and Decision-Focused Learning



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Topics

- 1. Introduction to Convex Optimization
- 2. Convex optimization problems
- 3. Decision-focused learning

Chapter 1



Introduction to Convex Optimization

What is optimization?

- Finding the best solution given some constraints
- Mathematical optimization

minimize $f_0(x)$

subject to $f_i(x) \leq b_i, \quad i = 1, \dots, m$

- $x = (x_1, \dots, x_n)$: optimization variables (or decision variables)
- $f_0: \mathbb{R}^n \rightarrow \mathbb{R}$: objective function
- $f_i: \mathbb{R}^n \rightarrow \mathbb{R}, i = 1, \dots, m$: constraint functions
- **Optimal solution x^*** has the smallest value of f_0 among all vectors that satisfy the constraints

What is optimization?

■ Examples

– Portfolio optimization

- Variables: amounts invested in different assets (stocks, bonds,...)
- Constraints: budget, max/min investment per asset, minimum required return, ...
- Objective: overall risk (e.g. return variance)

– Device sizing in electronic circuits

- Variables: device widths and lengths
- Constraints: manufacturing limits, timing requirements, maximum area
- Objective: power consumption

– Data fitting (or statistics, machine learning)

- Variables: model parameters
- Constraints: prior information, parameter limits
- Objective: measure of misfit or prediction error

What is optimization?

- Solving optimization problems
 - General optimization problems
 - Very difficult to solve
 - Methods involve some compromise
 - › E.g. very long computation time, or not always finding the solution
 - Exceptions
 - Certain problem classes can be solved efficiently and reliably
 - › Least-squares problems
 - › Linear programming problems
 - › **Convex optimization** problems

What is optimization?

- Comparison with machine learning (generally speaking)
 - Optimization is based on **mathematical modeling** of decisions and their consequences
 - Machine learning tries to **learn the mapping** between decisions and its consequences through data
- Are they completely different? No
 - Optimization needs domain knowledge and data to determine appropriate mathematical models and parameters
 - Machine learning may also take a modeling-based approach
 - Recently, reinforcement learning and sampling-based large-scale optimization techniques are very closely related
 - More importantly, “learning” involves optimization

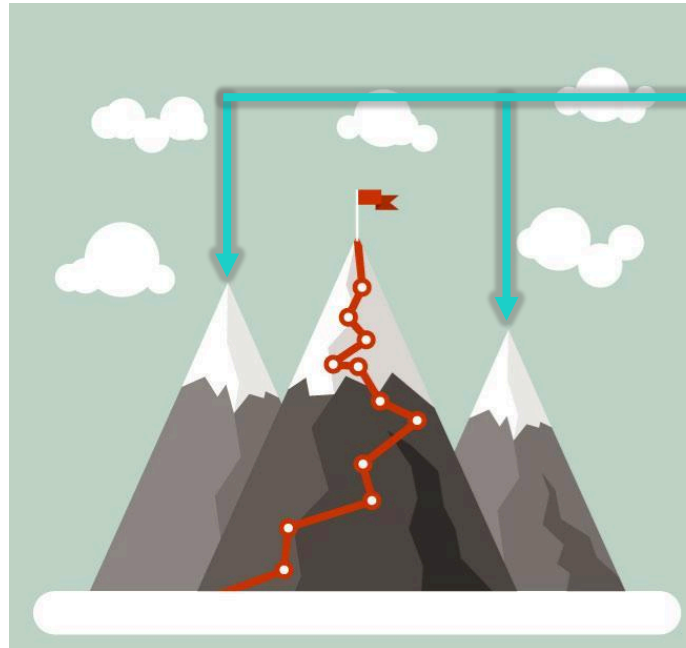
What is convex optimization?

- **General optimization** is like climbing a mountain
 - You are going to the *highest* peak
- A problem
 - There can be many peaks
 - You have to identify the highest peak before you climb
- A bigger problem
 - You are blindfolded
 - What can you do now? You are screwed!



What is convex optimization?

- A good news
 - For convex optimization problems, there is **only one peak**

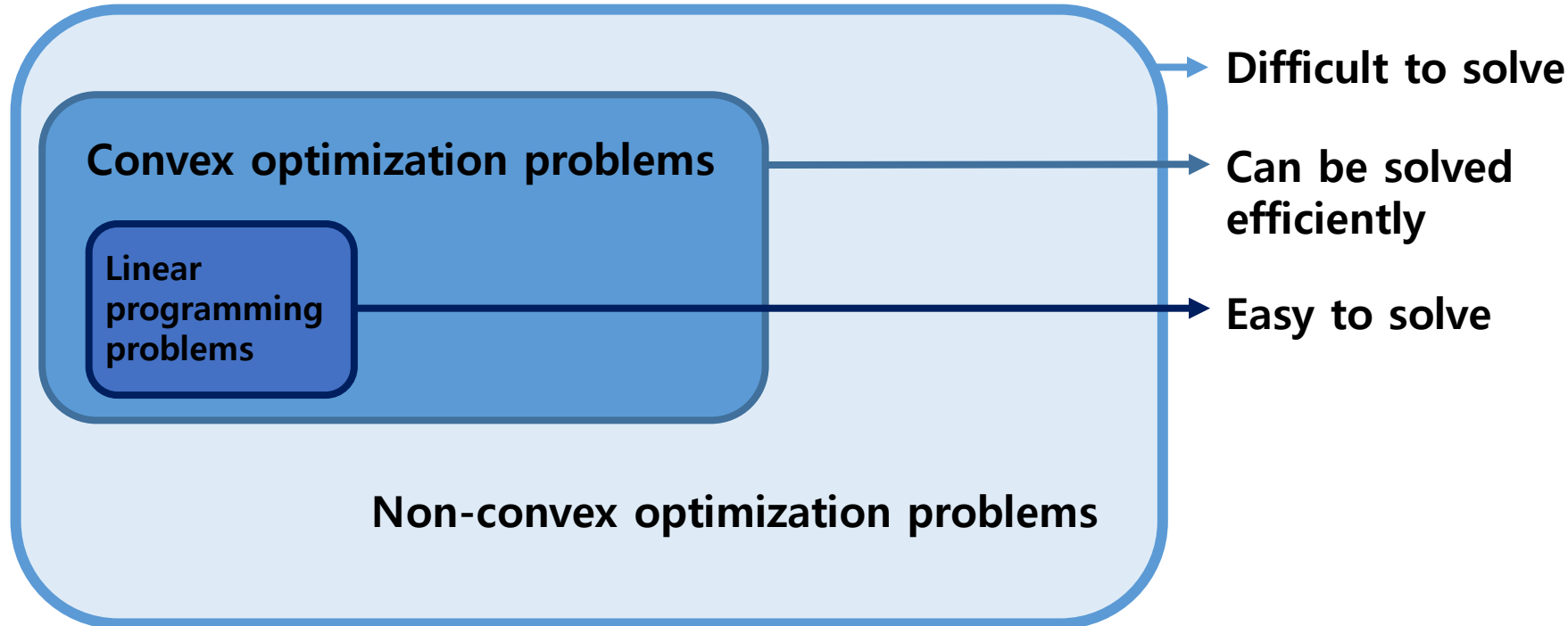


These don't exist
in convex optimization

- So, if you just go to a higher point for each step,
you will get to the peak eventually

What is convex optimization?

- General optimization problems



Least-squares

$$\text{minimize } \|Ax - b\|_2^2$$

- Solving least-square problems
 - Analytical solution: $x^* = (A^T A)^{-1} A^T b$
 - Reliable and efficient algorithms and software
 - Computation time proportional to $n^2 k$ ($A \in \mathbb{R}^{k \times n}$)
 - Less if structured
 - A mature technology
- Using least-squares
 - Least-squares problems are easy to recognize
 - A few standard techniques increase flexibility
 - e.g. including weights, adding regularization terms

Least-squares – An example

$$\text{minimize } \|X\boldsymbol{\beta} - \mathbf{y}\|_2^2$$

■ Polynomial regression

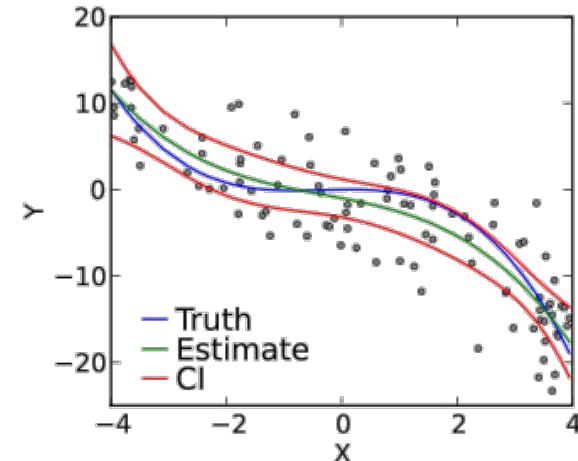
– Want to fit data $\{(x_i, y_i)\}_{i=1}^k$ to
$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_n x^n$$

– Then, we have

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_n x_i^n + \varepsilon_i,$$

– These can be written as follows

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_k \end{bmatrix} = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^n \\ 1 & x_2 & x_2^2 & \cdots & x_2^n \\ 1 & x_3 & x_3^2 & \cdots & x_3^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_k & x_k^2 & \cdots & x_k^n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_k \end{bmatrix}$$
$$\mathbf{y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}$$



$$i = 1, \dots, k$$

Linear programming

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, m\end{array}$$

- Solving linear programs
 - No analytical formula for solution
 - Reliable and efficient algorithms and software
 - Computation time proportional to n^2m if $m \geq n$
 - Less if structured
- Using linear programming
 - Not as easy to recognize as least squares problems
 - A few standard tricks to convert problems into linear programs
 - E.g. problems involving ℓ_1 - or ℓ_∞ -norms, piecewise-linear functions

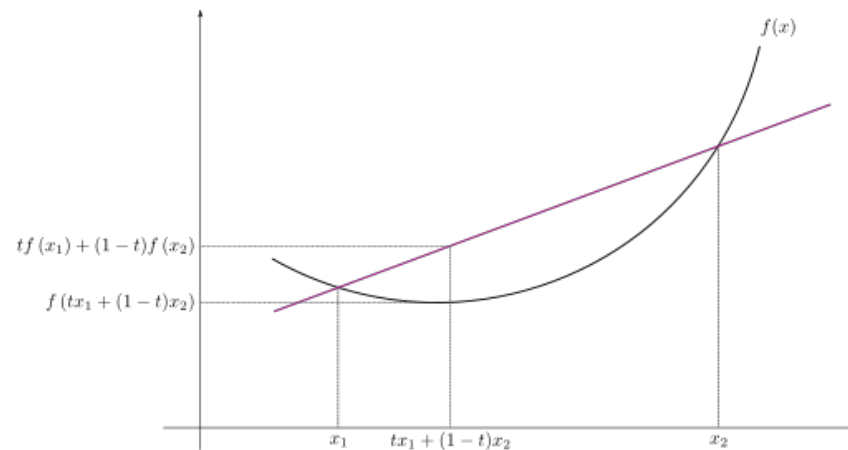
Convex optimization problem

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq b_i, \quad i = 1, \dots, m \end{aligned}$$

- Objective and constraint functions are convex
 - Function $f(x)$ is convex if and only if

$$f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2), \quad 0 \leq t \leq 1$$

Jensen's inequality



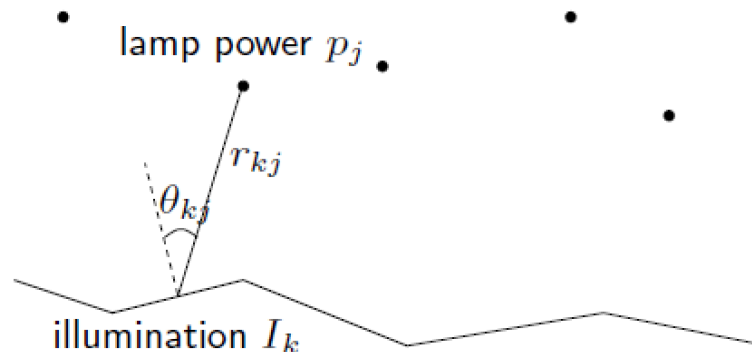
- Includes least-squares problems and linear programs as special cases

Convex optimization problem

- Solving convex optimization problems
 - No analytical solution
 - Reliable and efficient algorithms
 - Computation time (roughly) proportional to $\max\{n^3, n^2m, F\}$, where F is cost of evaluating f_i 's and their second derivatives
 - Almost a technology
- Using convex optimization
 - Often difficult to recognize
 - Many tricks for transforming problem into convex form
 - Surprisingly many problems can be solved via convex optimization

Example – Why formulation matters?

- m lamps illuminating n (small, flat) patches



- Intensity I_k at patch k depends linearly on lamp powers p_j :

$$I_k = \sum_{j=1}^m a_{kj} p_j, \quad a_{kj} = r_{kj}^{-2} \max\{\cos \theta_{kj}, 0\}$$

- **Problem:** achieve desired illumination I_{des} with bounded lamp powers

$$\begin{aligned} & \text{minimize} && \max_{k=1, \dots, n} |\log I_k - \log I_{des}| \\ & \text{subject to} && 0 \leq p_j \leq p_{\max}, \quad i = 1, \dots, m \end{aligned}$$

Example – Why formulation matters?

- How to solve?

1. Use uniform power: $p_j = p$, vary p

2. Use least-squares:

$$\text{minimize } \sum_{k=1}^n (I_k - I_{des})^2$$

round p_j if $p_j > p_{\max}$ or $p_j < 0$

3. Use weighted least-squares:

$$\text{minimize } \sum_{k=1}^n (I_k - I_{des})^2 + \sum_{j=1}^m w_j (p_j - p_{\max}/2)^2$$

iteratively adjust weights w_j until $0 \leq p_j \leq p_{\max}$

Example – Why formulation matters?

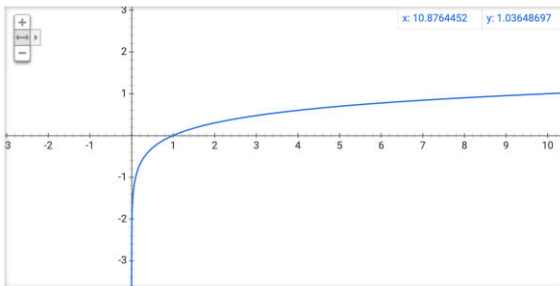
- How to solve?

- 4. Use linear programming:

$$\begin{aligned} &\text{minimize} && \max_{k=1,\dots,n} |I_k - I_{des}| \\ &\text{subject to} && 0 \leq p_j \leq p_{\max}, \quad j = 1, \dots, m \end{aligned}$$

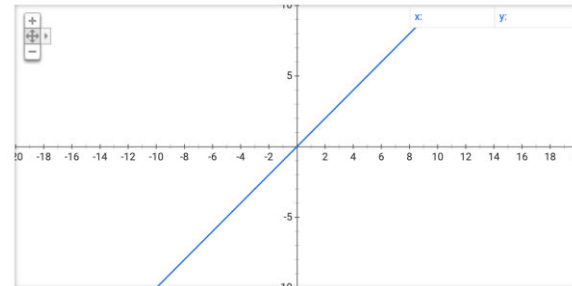
which can be solved via linear programming

Graph for $\log(x)$



VS

Graph for x



– Of course these are approximate (suboptimal) solutions

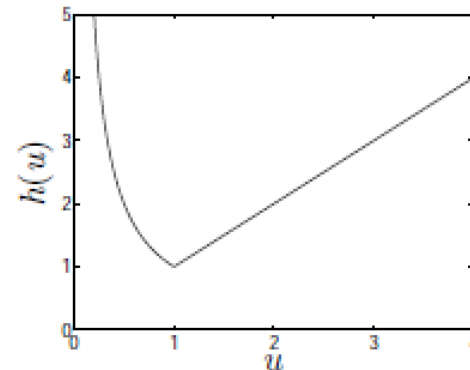
Example – Why formulation matters?

- How to solve?

5. Use **convex optimization**: problem is equivalent to

$$\begin{aligned} \text{minimize} \quad & f_0(p) = \max_{k=1,\dots,n} h(I_k/I_{des}) \\ \text{subject to} \quad & 0 \leq p_j \leq p_{\max}, \quad j = 1, \dots, m \end{aligned}$$

where $h(u) = \max\{u, 1/u\}$



- f_0 is convex because maximum of convex functions is convex
- **Exact** solution obtained with
effort $\approx$ modest factor $\times$ least-squares effort

Example – Why formulation matters?

- Additional constraints

- Does adding 1 or 2 below complicate the problem?

1. No more than half of total power is in any 10 lamps

2. No more than half of the lamps are on ($p_j > 0$)

- **Answer**

- With (1), it is still easy to solve
 - With (2), extremely difficult

- **Lessons**

- (untrained) **intuition doesn't always work**
 - Without the proper background, very easy problems can appear quite similar to very difficult problems

Chapter 2



Optimization Problems

Optimization problem in standard form

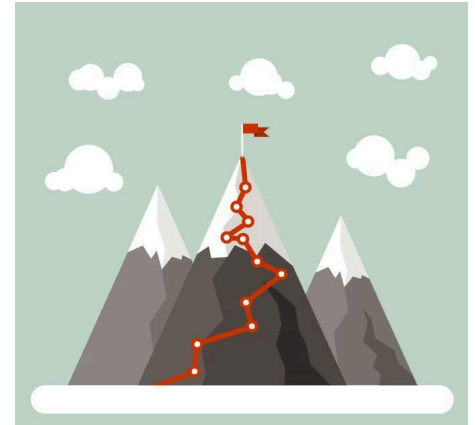
$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq 0, \quad i = 1, \dots, m \\ & && h_i(x) = 0, \quad i = 1, \dots, p \end{aligned}$$

- $x \in \mathbb{R}^n$ is the optimization variable (or decision variable)
- $f_0: \mathbb{R}^n \rightarrow \mathbb{R}$ is the objective or cost function
- $f_i: \mathbb{R}^n \rightarrow \mathbb{R}, i = 1, \dots, m$ are the inequality constraint functions
- $h_i: \mathbb{R}^n \rightarrow \mathbb{R}, i = 1, \dots, p$ are the equality constraint functions
- **Optimal value:**
 - $p^* = \inf \{f_0(x) \mid f_i(x) \leq 0, i = 1, \dots, m, h_i(x) = 0, i = 1, \dots, p\}$
 - $p^* = \infty$ if problem is infeasible (no x satisfies the constraints)
 - $p^* = -\infty$ if problem is unbounded below

Optimal and locally optimal points

- x is **feasible** if $x \in \mathbf{dom} f_0$ and it satisfies the constraints

- A feasible x is **optimal** if $f_0(x) = p^*$
 - X_{opt} is the set of optimal points



- x is **locally optimal** if there is an $R > 0$ s.t. x is optimal for

minimize $f_0(z)$
(over z)

subject to $f_i(z) \leq 0, \quad i = 1, \dots, m$

$h_i(z) = 0, \quad i = 1, \dots, p$

$\|z - x\|_2 \leq R$

Optimal and locally optimal points

■ Examples (with $n = 1, m = p = 0$, i.e., unconstrained)

– $f_0(x) = 1/x$, **dom** $f_0 = \mathbb{R}_{++}$

- $p^* = 0$, no optimal point

– $f_0(x) = -\log x$, **dom** $f_0 = \mathbb{R}_{++}$

- $p^* = -\infty$, unbounded below

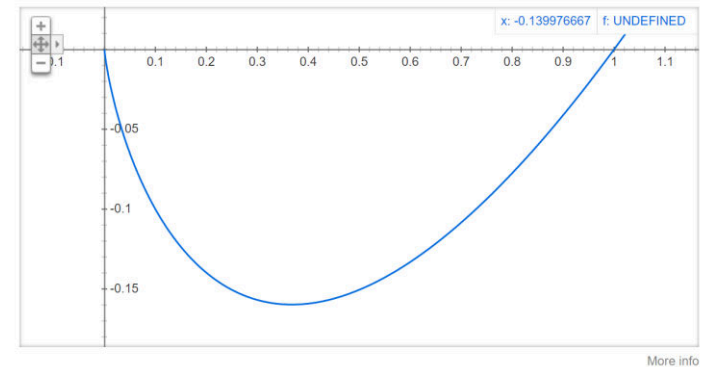
– $f_0(x) = x \log x$, **dom** $f_0 = \mathbb{R}_{++}$

- $p^* = -1/e$, $x = 1/e$ is optimal

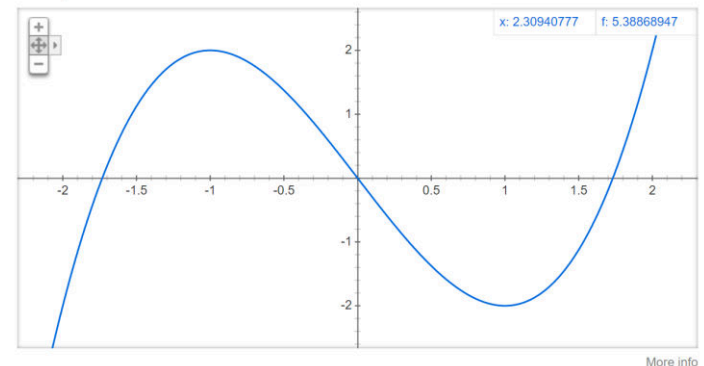
– $f_0(x) = x^3 - 3x$

- $p^* = -\infty$, local optimum at $x = 1$

Graph for $x \cdot \log(x)$



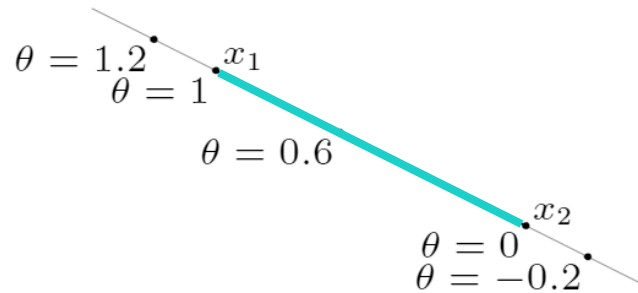
Graph for $x^3 - 3x$



Convex set

- **Line segment** between x_1 and x_2 : all points

$$x = \theta x_1 + (1 - \theta)x_2, \quad 0 \leq \theta \leq 1$$



- All convex combinations between x_1 and x_2

- **Convex set**: contains line segment between any two points in the set

$$x_1, x_2 \in C, \quad 0 \leq \theta \leq 1 \Rightarrow \theta x_1 + (1 - \theta)x_2 \in C$$

Convex set

■ Generalizing convex combination

– k points

- $x_1, \dots, x_k \in C \Rightarrow \theta_1 x_1 + \dots + \theta_k x_k \in C$
 - › where $\theta_1 + \dots + \theta_k = 1$ and $\theta_1, \dots, \theta_k \geq 0$

– Infinite sum

- $x_i \in C$ for $i = 1, 2, \dots \Rightarrow \sum_{i=1}^{\infty} \theta_i x_i \in C$
 - › where $\sum_{i=1}^{\infty} \theta_i = 1$ and $\theta_i \geq 0$ for $i = 1, 2, \dots$

– Integral

- $x \in C \Rightarrow \int_C p(x) x \, dx \in C$
 - › where $\int_C p(x) \, dx = 1$ and $p(x) \geq 0$ for all $x \in C$

– Probability distribution

- x is a random variable $\Rightarrow \mathbb{E}x \in C$

Convex function

- $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is **convex** if **dom** f is a convex set and

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

- for all $x, y \in \mathbf{dom} f$, $0 \leq \theta \leq 1$

← Jensen's inequality



- f is concave if $-f$ is convex
- f is strictly convex if **dom** f is convex and
$$f(\theta x + (1 - \theta)y) < \theta f(x) + (1 - \theta)f(y)$$
for $x, y \in \mathbf{dom} f$, $x \neq y$, $0 < \theta < 1$

Convex optimization problem

- Standard form **convex** optimization problem

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq 0, && i = 1, \dots, m \\ & && a_i^T x = b_i, && i = 1, \dots, p \end{aligned}$$

- $f_0, f_1, \dots, f_m$ are convex; equality constraints are affine

- Often written as

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq 0, && i = 1, \dots, m \\ & && Ax = b \end{aligned}$$

- **Important property:** feasible set of a convex optimization problem is convex

Convex optimization problem

▪ Example

$$\begin{aligned} &\text{minimize} && f_0(x) = x_1^2 + x_2^2 \\ &\text{subject to} && f_1(x) = x_1/(1 + x_2^2) \leq 0 \\ &&& h_1(x) = (x_1 + x_2)^2 = 0 \end{aligned}$$

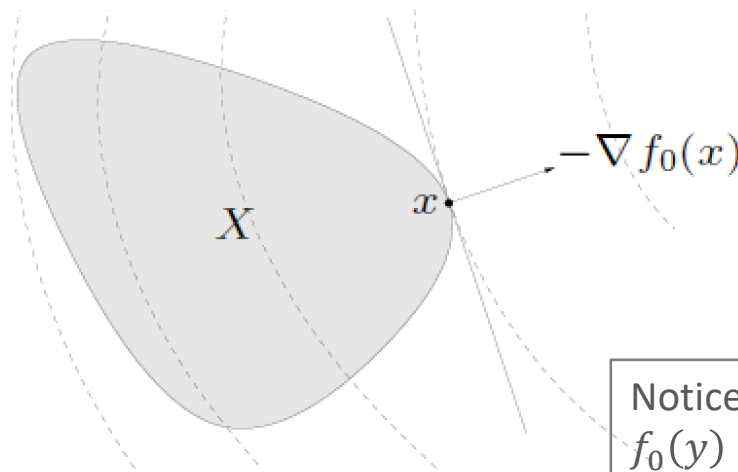
- f_0 is convex; feasible set $\{(x_1, x_2) \mid x_1 = -x_2 \leq 0\}$ is convex
- But, not a convex problem: f_1 non-convex, h_1 non-affine
- Equivalent (but not identical) to the following convex problem

$$\begin{aligned} &\text{minimize} && x_1^2 + x_2^2 \\ &\text{subject to} && x_1 \leq 0 \\ &&& x_1 + x_2 = 0 \end{aligned}$$

Optimality criterion for differentiable f_0

- x is **optimal** if and only if it is feasible and

$$\nabla f_0(x)^T (y - x) \geq 0 \quad \text{for all feasible } y$$



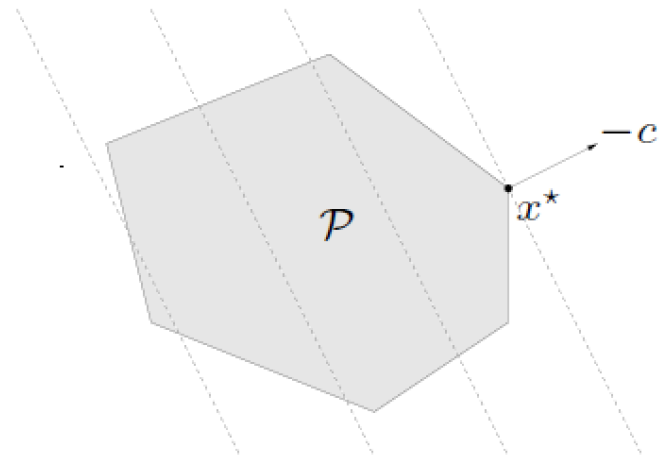
Notice that for all $x, y \in \text{dom } f_0$,
 $f_0(y) \geq f_0(x) + \nabla f_0(x)^T (y - x)$.
 x being optimal means $f_0(y) \geq f_0(x)$

- If nonzero, $-\nabla f_0(x)$ defines a supporting hyperplane to feasible set X at x

Linear program (LP)

$$\begin{aligned} & \text{minimize} && c^T x + d \\ & \text{subject to} && Gx \leq h \\ & && Ax = b \end{aligned}$$

- with $G \in \mathbb{R}^{m \times n}$ and $A \in \mathbb{R}^{p \times n}$
- **Convex** problem with affine objective and constraint functions
- Feasible set is a polyhedron



Linear program (LP)

■ Examples

– Diet problem

- Choose quantities $x_1, \dots, x_n$ of n foods
- One unit of food j costs c_j , contains amount a_{ij} of nutrient i
- Healthy diet requires nutrient i in quantity at least b_i
- To find cheapest healthy diet, solve

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax \geq b, x \geq 0\end{array}$$

Linear program (LP)

■ Examples

– Piecewise-linear minimization

- Minimizing a piecewise-linear, convex function

$$\text{minimize} \quad \max_{i=1,\dots,m} (a_i^T x + b_i)$$

- is equivalent to the following epigraph problem

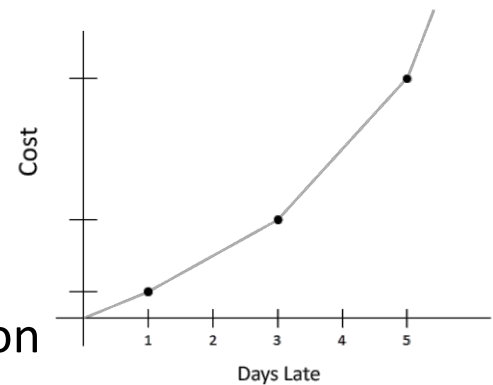
$$\text{minimize} \quad t$$

$$\text{subject to} \quad \max_{i=1,\dots,m} (a_i^T x + b_i) \leq t$$

- can be expressed using m separate inequalities

$$\text{minimize} \quad t$$

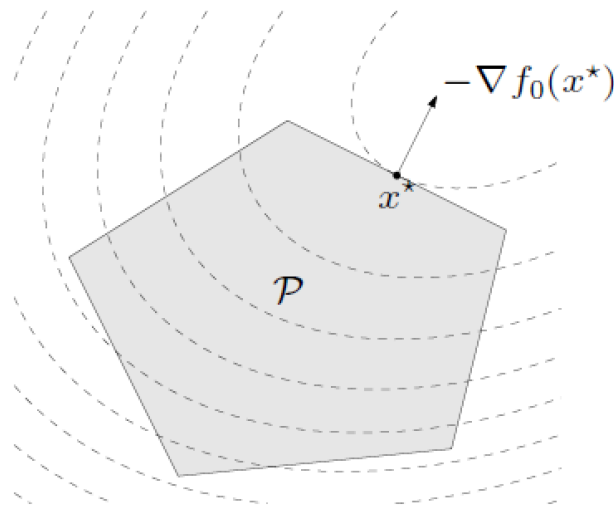
$$\text{subject to} \quad a_i^T x + b_i \leq t, \quad i = 1, \dots, m$$



Quadratic program (QP)

$$\begin{aligned} & \text{minimize} && \frac{1}{2}x^T Px + q^T x + r \\ & \text{subject to} && Gx \leq h \\ & && Ax = b \end{aligned}$$

- $P \in \mathbf{S}_+^n$, so objective is convex quadratic
- Minimize a convex quadratic function over a polyhedron



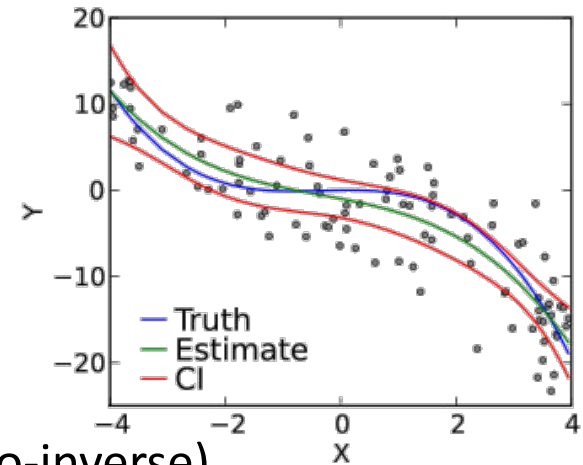
Quadratic program (QP)

■ Examples

– Least-squares and regression

$$\text{minimize } \|Ax - b\|_2^2$$

- Analytical solution $x^* = A^\dagger b$ ($A^\dagger$ is pseudo-inverse)
- Can add linear constraints (e.g. $l \leq x \leq u$)



– Linear program with random cost

(c is random with mean $\bar{c}$ and covariance Σ)

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Gx \leq h \\ & Ax = b \end{array} \quad \longrightarrow \quad \begin{array}{ll} \text{minimize} & \mathbb{E}[c^T x] + \gamma \mathbf{var}(c^T x) \\ \text{subject to} & Gx \leq h \\ & Ax = b \end{array}$$

- $\mathbb{E}[c^T x] = \bar{c}^T x$ and $\mathbf{var}(c^T x) = x^T \Sigma x$
- $\gamma > 0$ is risk-aversion parameter
(controls trade-off between expected cost and variance (risk))

Quadratic program (QP)

■ Examples

– Markowitz portfolio optimization

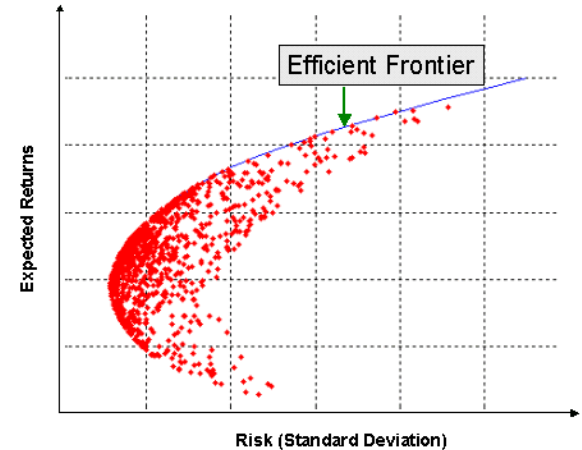
- n number of investment assets
 - › r_i = rate of return of asset i
 - › $r = [r_1 \ r_2 \ \dots \ r_n]$ is random with mean μ and covariance Σ
- Choose investment proportions $w_1, \dots, w_n$ on n assets
 - › $w_1 + \dots + w_n = 1$ (budget constraint)
 - › $w_1, \dots, w_n \geq 0$ (no short constraint) (can be omitted in some cases)
- We want to minimize risk and maximize return

$$\text{minimize} \quad w^T \Sigma w - \lambda \mu^T w$$

$$\text{subject to} \quad \mathbb{1}^T w = 1$$

$$w \geq 0$$

- › $\lambda > 0$ is risk-preference parameter



Quadratically constrained quadratic program (QCQP)

$$\begin{aligned} & \text{minimize} && \frac{1}{2}x^T P_0 x + q_0^T x + r_0 \\ & \text{subject to} && \frac{1}{2}x^T P_i x + q_i^T x + r_i \leq 0, \quad i = 1, \dots, m \\ & && Ax = b \end{aligned}$$

- $P_i \in \mathbf{S}_{+}^n$; hence, objective and inequality constraints are convex quadratic
- If $P_1, \dots, P_m \in \mathbf{S}_{++}^n$, feasible region is intersection of m ellipsoids and an affine set
- QP include LP as a special case ($P = 0$)
- QCQP include QP as a special case ($P_i = 0$ for all $i > 0$)
- Hence, $\text{LP} \subset \text{QP} \subset \text{QCQP}$

Quadratically constrained quadratic program (QCQP)

■ Sharpe ratio maximization

$$\text{maximize}_w \frac{w^T \mu - r_f}{\sqrt{w^T \Sigma w}}$$

$$\text{subject to } \mathbf{1}^T w = 1$$



$$\text{maximize}_{y, t} y^T \mu - t r_f$$

$$\text{subject to } \mathbf{1}^T y = t$$

$$\sqrt{y^T \Sigma y} \leq 1$$

$$t \geq 0$$

Sharpe ratio

$$\text{Sharpe ratio} = \frac{\mu_p - r_f}{\sigma_p} = \frac{w^T \mu - r_f}{\sqrt{w^T \Sigma w}}$$

A measure of return over risk

- It gives the same result ($w = y/t$)
- This is a quadratically-constrained quadratic programming (QCQP) problem, which is convex

Second-order cone program (SOCP)

$$\begin{aligned} & \text{minimize} && f^T x + r_0 \\ & \text{subject to} && \|A_i x + b_i\|_2 \leq c_i^T x + d_i, \quad i = 1, \dots, m \\ & && Fx = g \end{aligned}$$

- With $A_i \in \mathbb{R}^{n_i \times n}$, $b_i \in \mathbb{R}^{n_i}$, $F \in \mathbb{R}^{p \times n}$
- Inequalities are called **second-order cone (SOC) constraints**:
 $(A_i x + b_i, c_i^T x + d_i) \in \text{second-order cone in } \mathbb{R}^{n_i+1}$
- For $n_i = 0$, reduces to an LP; if $c_i = 0$, reduces to a QCQP
- Hence, more general than QCQP and LP

Second-order cone program (SOCP)

▪ Example

– Robust linear programming

- Parameters in optimization problems are often **uncertain**
- For example, in LP

$$\text{minimize } c^T x$$

$$\text{subject to } a_i^T x \leq b_i, \quad i = 1, \dots, m,$$

there can be **uncertainty in c, a_i, b_i**

- **Two common approaches** to handling uncertainty (in a_i , for simplicity)
 - › **Deterministic model:** constraints must hold for all $a_i \in \mathcal{E}_i$

$$\text{minimize } c^T x$$

$$\text{subject to } a_i^T x \leq b_i, \quad \text{for all } a_i \in \mathcal{E}_i, \quad i = 1, \dots, m$$

- › **Stochastic model:** a_i is random; constraints must hold with probability η

$$\text{minimize } c^T x$$

$$\text{subject to } \mathbf{prob}(a_i^T x \leq b_i) \geq \eta, \quad i = 1, \dots, m$$

Second-order cone program (SOCP)

■ Example

– Robust linear programming (cont'd)

• Deterministic approach via SOCP

- › Choose an ellipsoid as $\mathcal{E}_i$
 $\mathcal{E}_i = \{\bar{a}_i + P_i u \mid \|u\|_2 \leq 1\} \quad (\bar{a}_i \in \mathbb{R}^n, P_i \in \mathbb{R}^{n \times n})$

› Robust LP

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && a_i^T x \leq b_i, \text{ for all } a_i \in \mathcal{E}_i, \ i = 1, \dots, m \end{aligned}$$

» is equivalent to the SOCP

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && \bar{a}_i^T x + \|P_i^T x\|_2 \leq b_i, \ i = 1, \dots, m \end{aligned}$$

» Since $\sup_{\|u\|_2 \leq 1} (\bar{a}_i + P_i u)^T x = \bar{a}_i^T x + \|P_i^T x\|_2$

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && a_i^T x \leq b_i, \ i = 1, \dots, m \end{aligned}$$

Second-order cone program (SOCP)

■ Example

– Robust linear programming (cont'd)

• Stochastic approach via SOCP

- › Assume a_i is Gaussian with mean $\bar{a}_i$, covariance Σ_i ($a_i \sim \mathcal{N}(\bar{a}_i, \Sigma_i)$)
- › $a_i^T x$ is a Gaussian r.v. with mean $\bar{a}_i^T x$, variance $x^T \Sigma_i x$; hence

$$\text{prob}(a_i^T x \leq b_i) = \Phi\left(\frac{b_i - \bar{a}_i^T x}{\|\Sigma_i^{1/2} x\|_2}\right)$$

where $\Phi(x) = (1/\sqrt{2\pi}) \int_{-\infty}^x e^{-t^2/2} dt$ is CDF of $\mathcal{N}(0,1)$

- › **Robust LP**
minimize $c^T x$
subject to $\text{prob}(a_i^T x \leq b_i) \geq \eta, i = 1, \dots, m$

- » With $\eta \geq 1/2$ is equivalent to the SOCP
minimize $c^T x$

$$\text{subject to } \bar{a}_i^T x + \Phi^{-1}(\eta) \|\Sigma_i^{1/2} x\|_2 \leq b_i, i = 1, \dots, m$$

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, i = 1, \dots, m \end{array}$$

Semidefinite program (SDP)

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && x_1 F_1 + x_2 F_2 + \cdots + x_n F_n + G \preceq 0 \\ & && Ax = b \end{aligned}$$

– $G, F_1, \dots, F_n \in \mathbf{S}^k$ and $A \in \mathbb{R}^{p \times n}$

– Inequality constraint is called linear matrix inequality (LMI)

– Includes problems with multiple LMI constraints: for example,

$$x_1 \hat{F}_1 + \cdots + x_n \hat{F}_n + G \preceq 0, \quad x_1 \tilde{F}_1 + \cdots + x_n \tilde{F}_n + G \preceq 0$$

is equivalent to single LMI

$$x_1 \begin{bmatrix} \hat{F}_1 & 0 \\ 0 & \tilde{F}_1 \end{bmatrix} + x_2 \begin{bmatrix} \hat{F}_2 & 0 \\ 0 & \tilde{F}_2 \end{bmatrix} + \cdots + x_n \begin{bmatrix} \hat{F}_n & 0 \\ 0 & \tilde{F}_n \end{bmatrix} + \begin{bmatrix} \hat{G} & 0 \\ 0 & \tilde{G} \end{bmatrix} \preceq 0$$

Semidefinite program (SDP)

■ Standard form semidefinite program

$$\begin{array}{ll}\text{minimize} & \mathbf{tr}(CX) \\ \text{subject to} & \mathbf{tr}(A_i X) = b_i, \quad i = 1, \dots, p \\ & X \succeq 0\end{array}$$

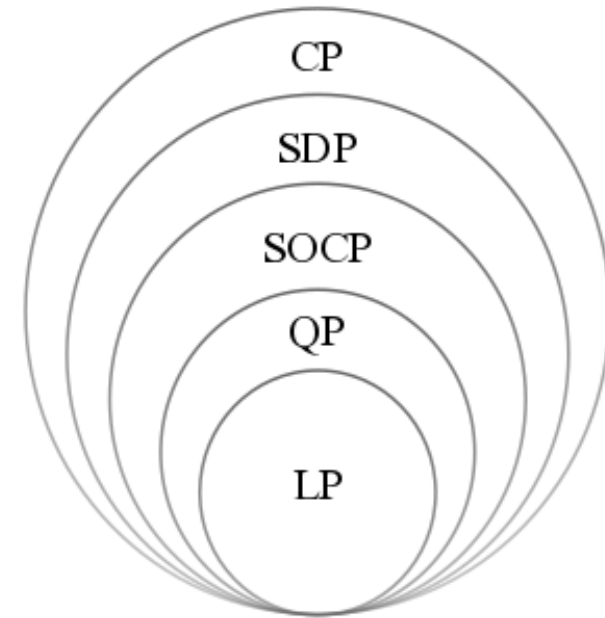
$$- C, A_1, \dots, A_p \in \mathbf{S}^n$$

– Note that $\mathbf{tr}(CX) = \sum_{i,j=1}^n C_{ij}X_{ij}$ is the form of a general real-valued linear function on $\mathbf{S}^n$

- Hence, it can be regarded as a matrix version of LP

Hierarchy of convex optimization problems

- CP: cone program
- SDP: Semi-definite program
- SOCP: Second-order cone program
- QP: quadratic program
- LP: linear program



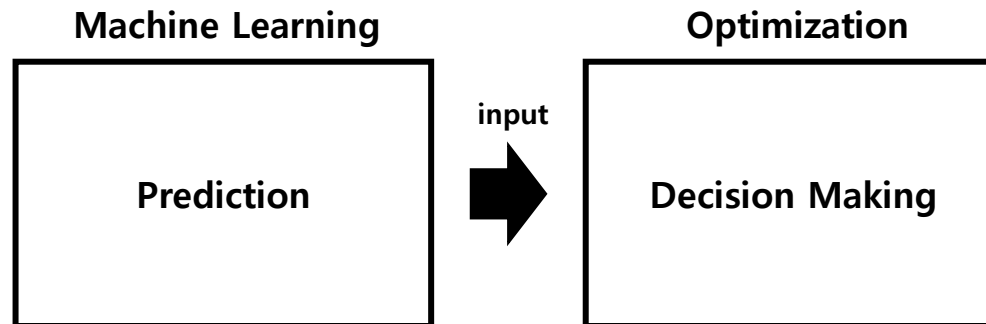
Chapter 3



Decision-focused learning

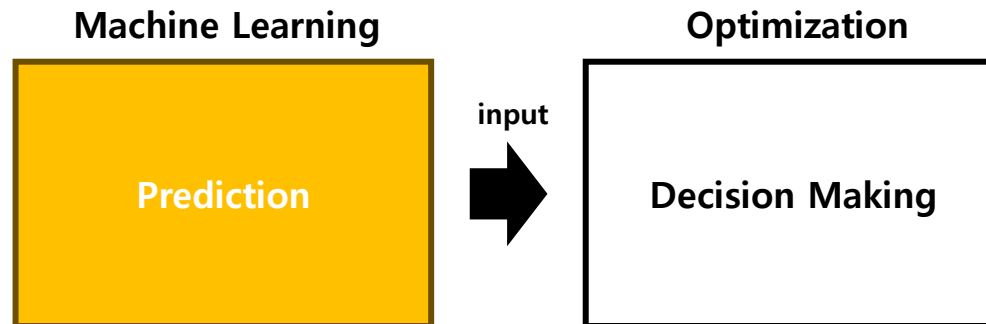
ML-based decision making process

- Issues in usual ML-based decision making process
 - Predict-then-optimize



ML-based decision making process

- Issues in usual ML-based decision making process
 - Predict-then-optimize

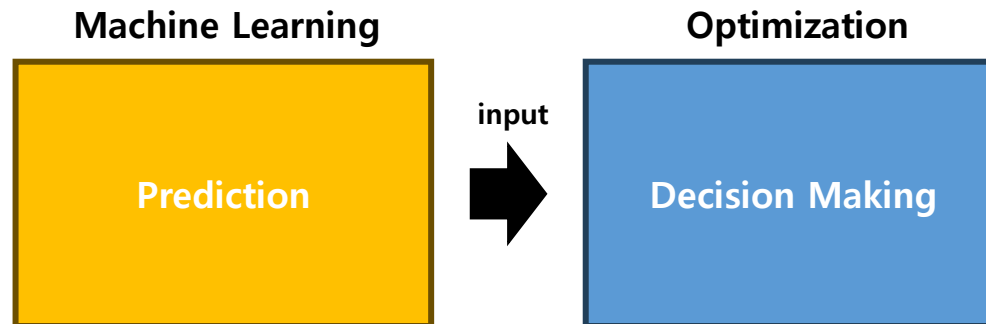


Issue 1

Predictions should contain errors
(Garbage in garbage out)

ML-based decision making process

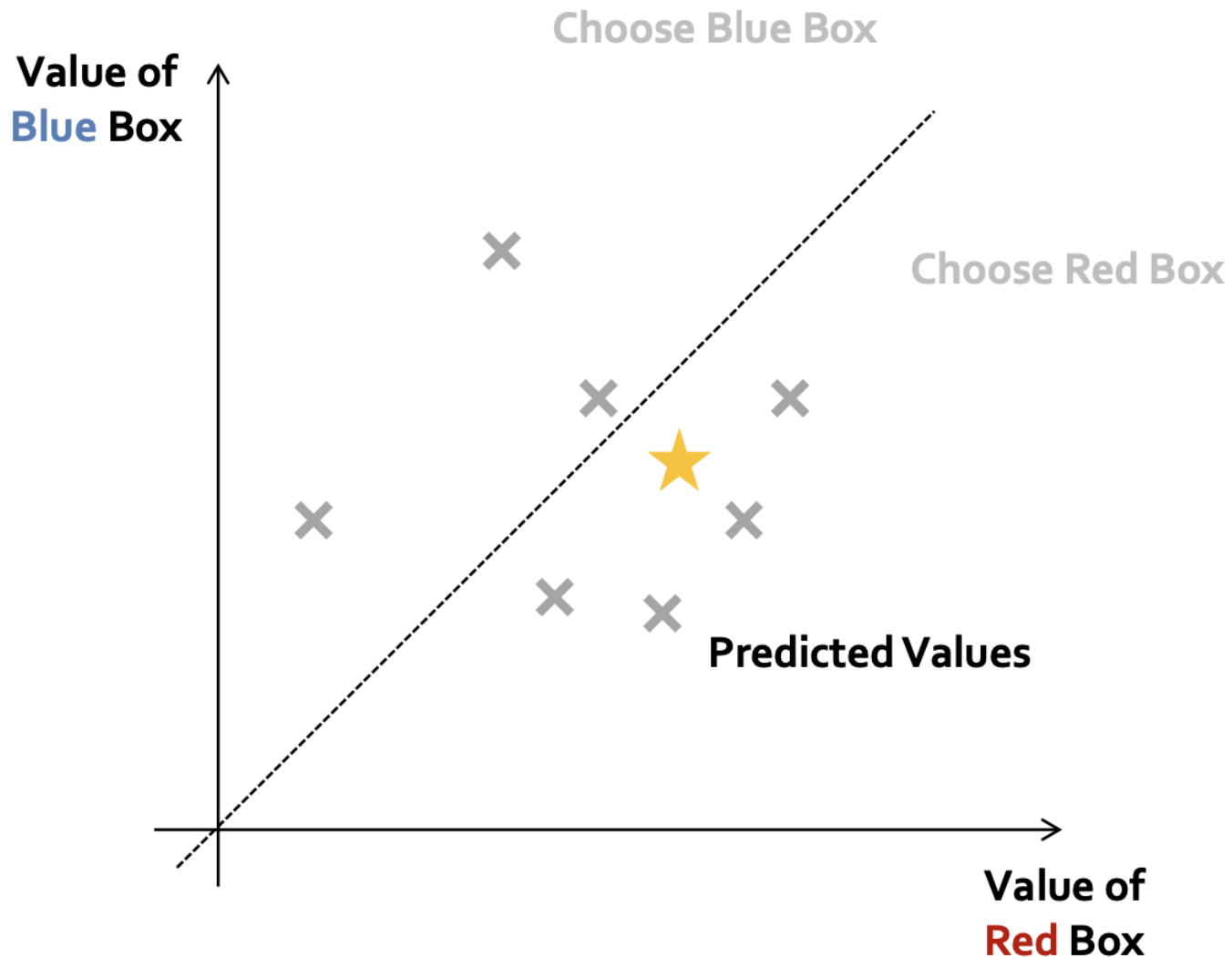
- Issues in usual ML-based decision making process
 - Predict-then-optimize



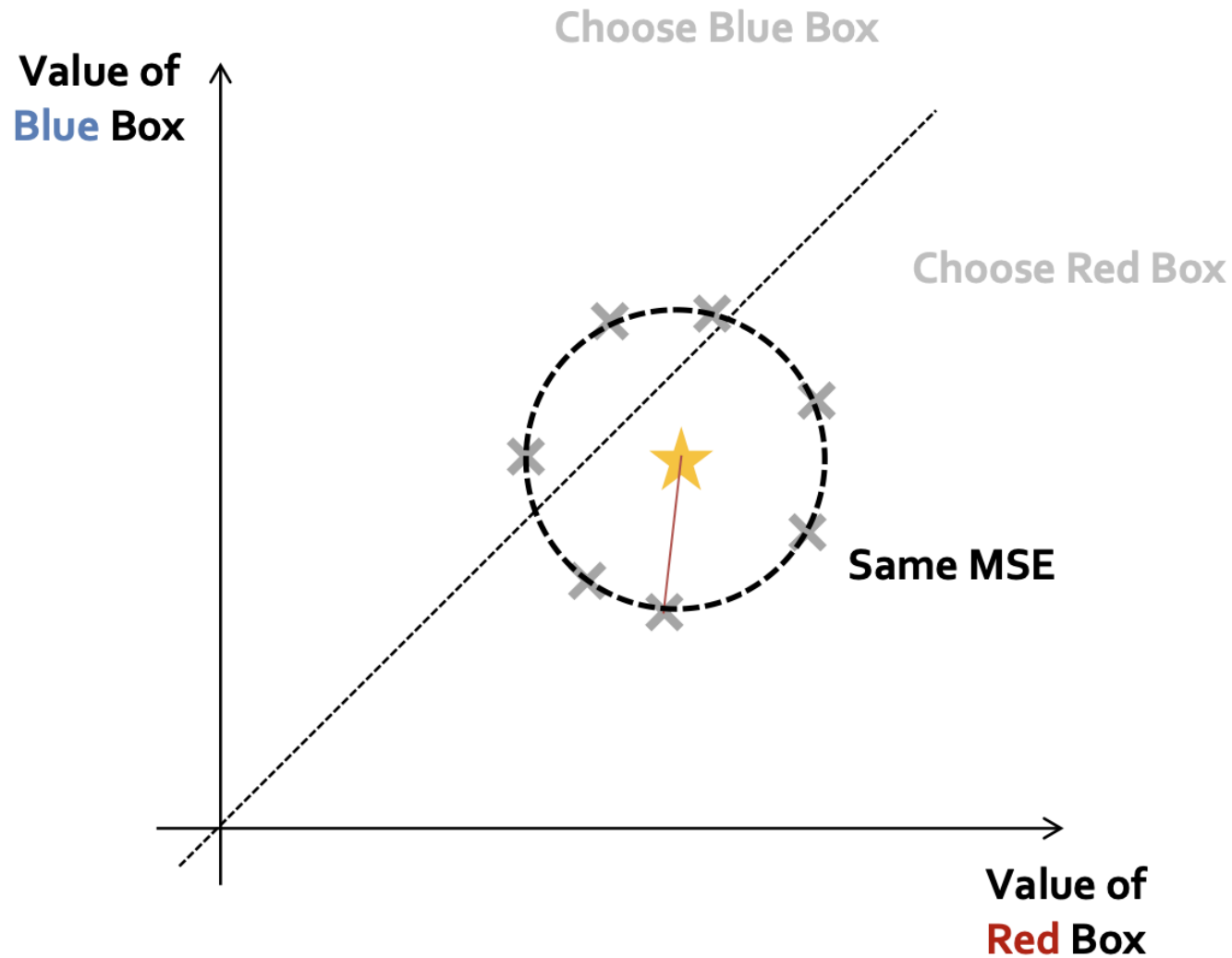
Issue 2

Best prediction $\neq$ Optimal decision

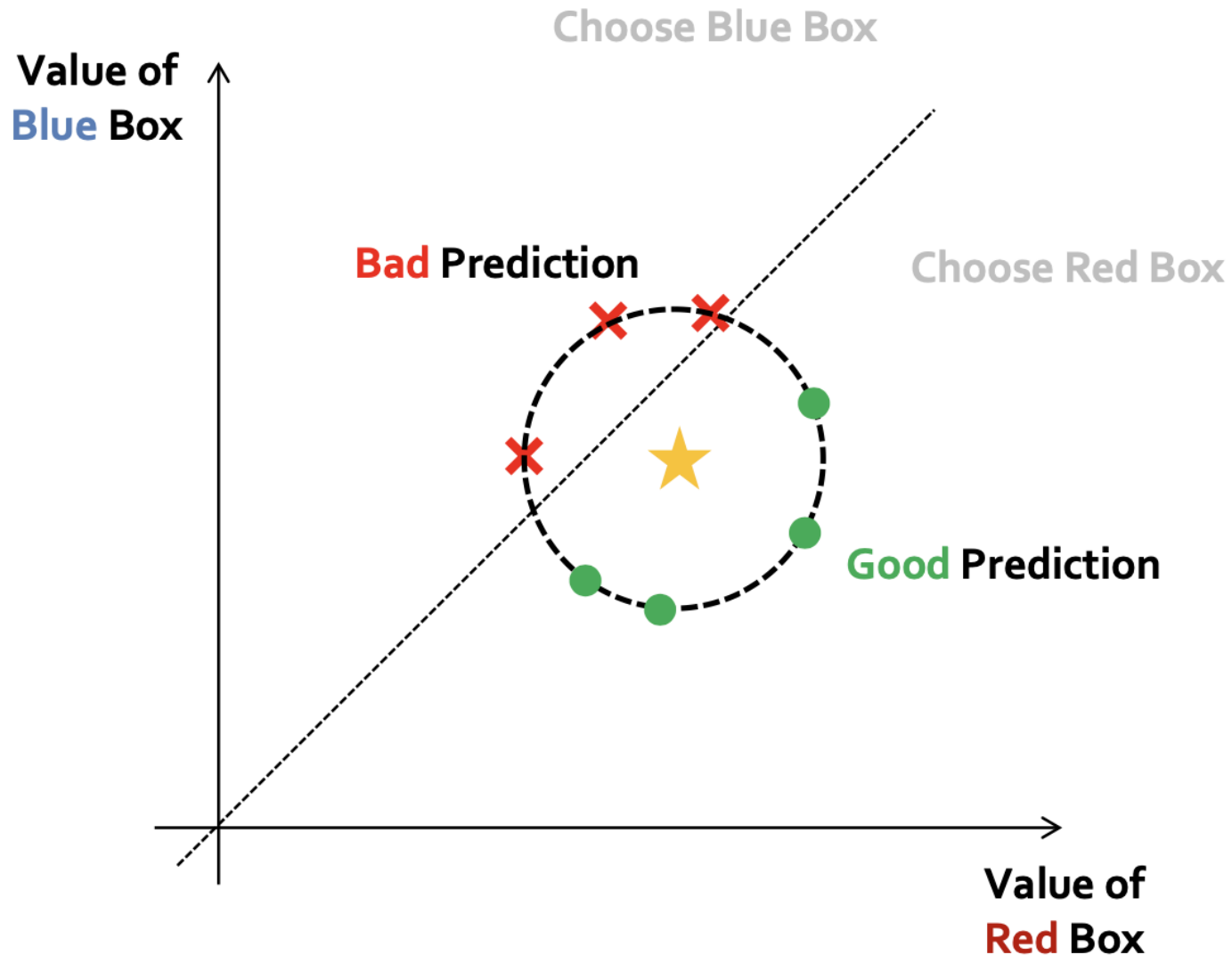
Motivating examples



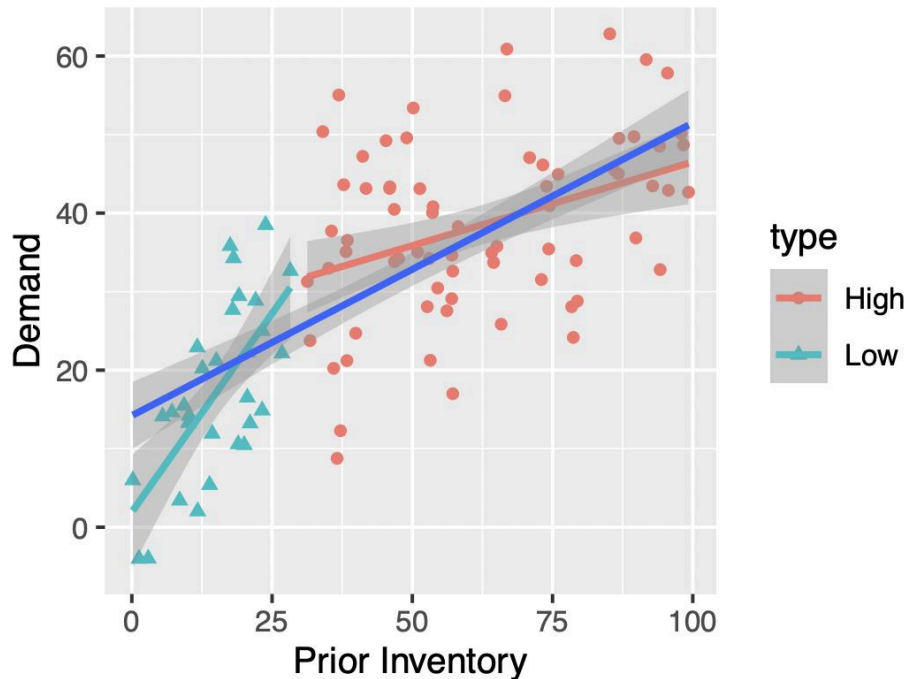
Motivating examples



Motivating examples



Motivating examples

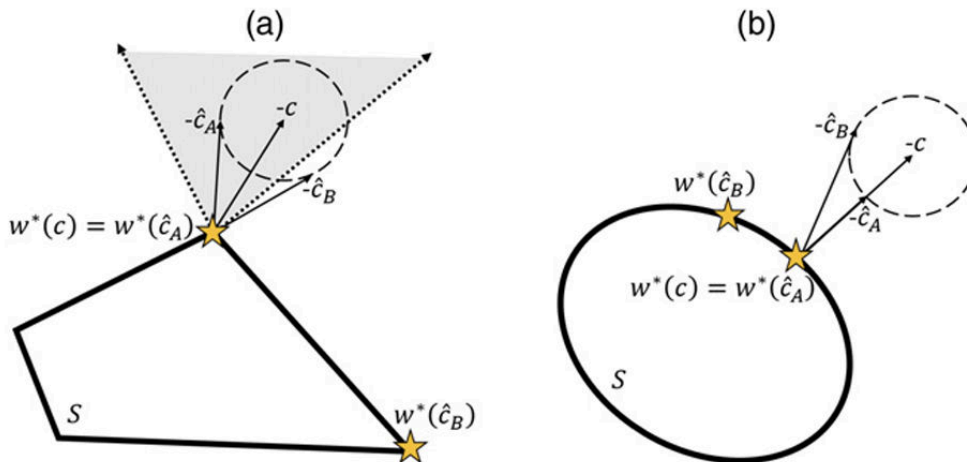


- Two different facilities
 - High prior inventory
 - In most cases, prior inventory is higher than demand
 - No action required
 - Low prior inventory
 - In most cases, demand is higher than prior inventory
 - Need to do something

But MSE-minimizing prediction model cannot reflect this

Motivating examples

Figure 1. Geometric Illustration of SPO Loss



Notes. (a) Polyhedral feasible region. (b) Elliptic feasible region.

- Minimize $c^T x$ within a feasible region
 - (a) predictions $\hat{c}_A$ and $\hat{c}_B$ have similar level of error
 - But, for decisions, $w^*(\hat{c}_A) = w^*(c)$, while $w^*(\hat{c}_B)$ is very different
 - (b) predictions $\hat{c}_A$ and $\hat{c}_B$ have similar level of error
 - But here, $\hat{c}_A$ has the same direction with c and $\hat{c}_B$ does not
 - For decisions, $w^*(\hat{c}_A) = w^*(c)$, while $w^*(\hat{c}_B)$ is quite different

Decision-focused learning

- $\mathcal{D} := \{(x_1, y_1), \dots, (x_n, y_n)\}$, a decision should be made given x
- Predict then optimize:

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \sum_{i=1}^n \mathcal{L}^{pred}(f_{\theta}(x_i), y_i) \quad \text{e.g., } \mathcal{L}^{pred}(\hat{y}, y) = (y - \hat{y})^2$$

$$A^{\pi}(x) = a^{\star}(f_{\hat{\theta}}(x)) \quad \text{where } a^{\star}(y) := \arg \min_{a \in \mathcal{A}} c(a, y)$$

- Decision-focused learning

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \sum_{i=1}^n \mathcal{L}^{dec}(f_{\theta}(x_i), y_i)$$

$$\mathcal{L}^{dec}(\hat{y}, y) := c(a^{\star}(\hat{y}), y)$$

Decision-focused learning

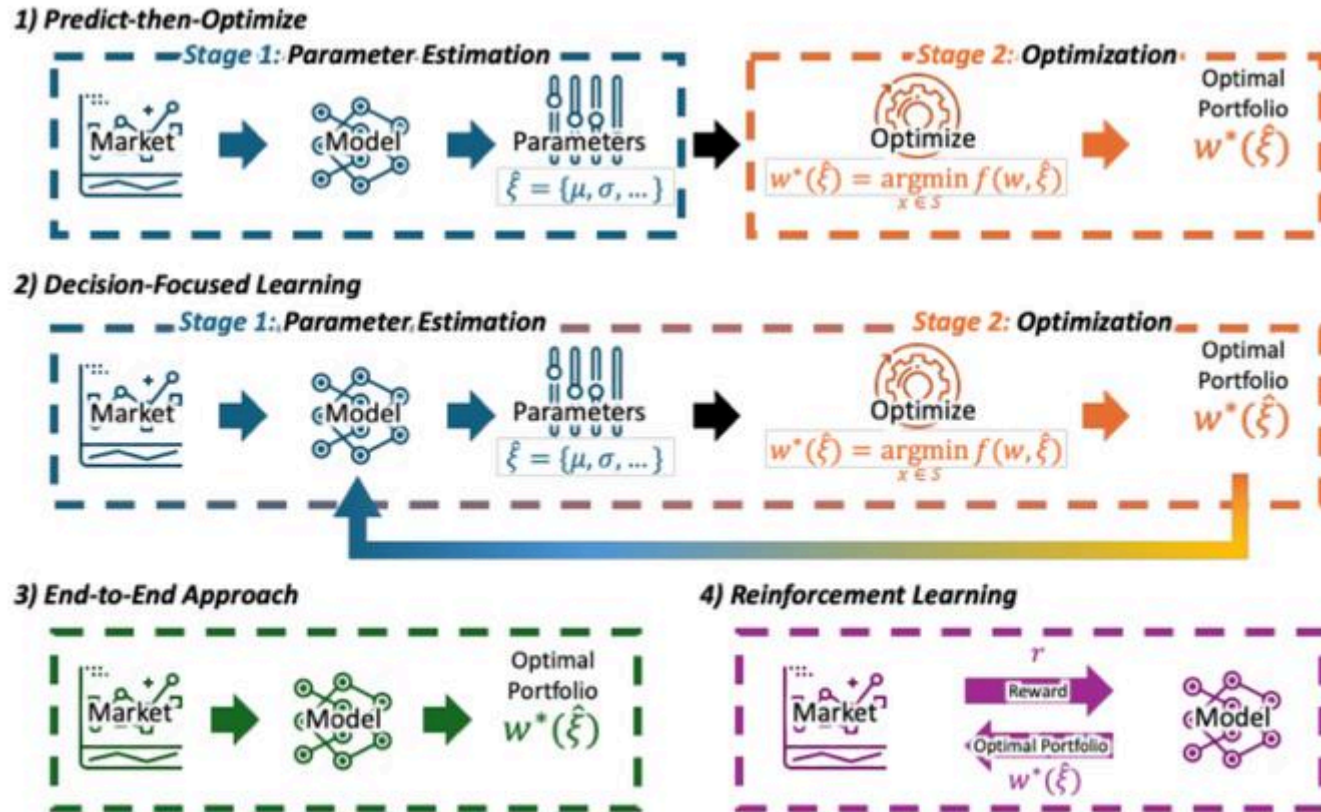


Exhibit 2. Graphical illustration of predict-then-optimize, decision-focused learning, end-to-end approach, and reinforcement learning for portfolio optimization

Lee, Yongjae; Kim, Jang Ho; Kim, Woo Chang; Fabozzi, Frank J. (2024) "An Overview of Machine Learning for Portfolio Optimization," *Journal of Portfolio Management*, 51(2), 131-148

Decision-focused learning

$$\min_{\theta \in \Theta} \sum_{i=1}^n \mathcal{L}^{dec}(f_{\theta}(x_i), y_i) \quad \mathcal{L}^{dec}(\hat{y}, y) := c(a^*(\hat{y}), y)$$

- If everything is differentiable,

$$\frac{d\mathcal{L}^{dec}}{d\theta} = \frac{dc(a^*(f_{\theta}(x_i)), y_i)}{da^*(f_{\theta}(x_i))} \cdot \frac{da^*(f_{\theta}(x_i))}{df_{\theta}(x_i)} \cdot \frac{df_{\theta}(x_i)}{d\theta}$$

- The second term $\frac{da^*(y)}{dy}$ is particularly tricky
 - $a^*(y)$ may not be uniquely defined or differentiable

Decision-focused learning

■ Proxy Objectives (Surrogates)

- In many cases, it is computationally challenging to minimize the decision loss
 - involves an inner optimization problem, $a^*(f_\theta(x_i))$
 - may not be differentiable
- People have suggested to use proxy-objectives (surrogates)

$$\min_{\theta \in \Theta} \sum_{i=1}^n \mathcal{L}^{dec}(f_\theta(x_i), y_i) \quad \Rightarrow \quad \min_{\theta \in \Theta} \sum_{i=1}^n \hat{\mathcal{L}}^{dec}(f_\theta(x_i), y_i)$$

- Linear reparameterization by Wang et al. (2020)
- SPO+ (Elmachtoub and Grigas, 2022)
- Local approximation (Chung et al., 2022, Shah et al., 2022)

Decision-focused learning

- Proxy Objectives (Surrogates)
 - Locally optimized decision losses (LODL) by Shah et al. (2022)

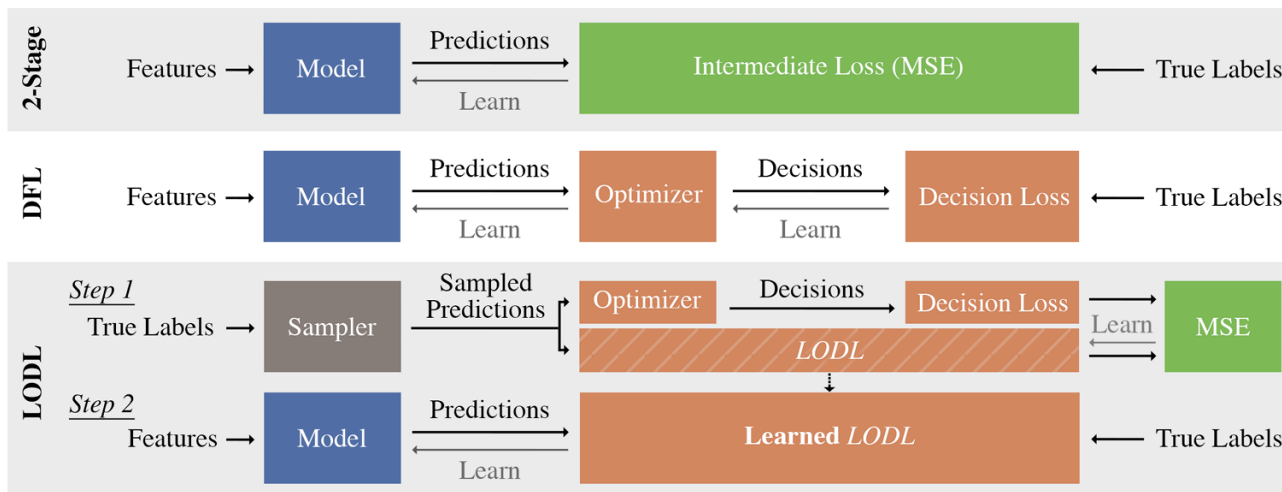


Figure 1: Schematic highlighting how the predictive model M_θ is learned in different approaches. LODL does not require backpropagating through the optimization problem.

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_{n=1}^N DL(M_\theta(x_n), y_n)$$

$$LODL_{\phi_i}(\hat{y}_i) \approx DL(\hat{y}_i, y_i) - DL(y_i, y_i) \implies DL(\hat{y}_i, y_i) \approx LODL_{\phi_i}(\hat{y}_i) + \overbrace{DL(y_i, y_i)}^{\text{constant}}$$

Decision-focused learning

■ Tangent-Space Projection

– Prediction Error As Regret-gradient (PEAR) by Lee et al. (2026)

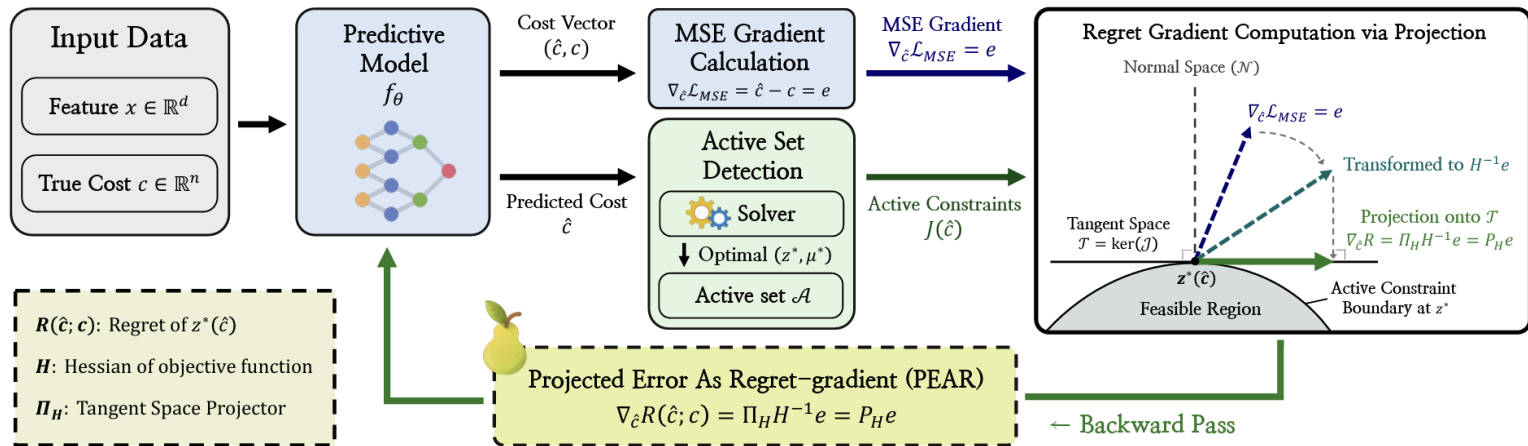


Figure 1. **The PEAR framework.** PEAR provides a direct pipeline for transforming prediction error into a regret gradient via a projection onto the local decision geometry. After identifying the active constraints, PEAR rescales the error by local curvature and projects it onto the feasible tangent space, filtering out decision-irrelevant components.

$$\nabla_{\hat{c}} \mathcal{R}(\hat{c}; c) = \underbrace{(I - H^{-1} J^\top (J H^{-1} J^\top)^{-1} J)}_{\text{tangent projector}} \underbrace{(H^{-1})}_{\text{curvature rescaler}} e,$$

Lee, J., Jin, S., & Lee, Y. (2026). Decision-Focused Learning via Tangent-Space Projection of Prediction Error. *Forty-Third International Conference on Machine Learning (ICML 2026)*

Decision-focused learning

■ Tangent-Space Projection

– Prediction Error As Regret-gradient (PEAR) by Lee et al. (2026)

Table 1. **LP benchmarks.** Normalized regret (% , ↓) and training time (s, ↓), over 5 seeds. Best in **bold** and second best underlined.

Task	Method	Deg 2		Deg 4		Deg 6		Deg 8	
		Regret(%)	Time(s)	Regret(%)	Time(s)	Regret(%)	Time(s)	Regret(%)	Time(s)
Shortest Path	MSE	0.135±0.037	7.5±0.8	1.967±0.565	7.5±2.1	6.561±1.514	8.9±1.5	15.428±3.085	6.1±0.1
	SPO+	0.120±0.046	48.7±11.6	0.761±0.242	41.8±10.8	<u>2.281±0.935</u>	44.1±5.4	<u>4.403±1.491</u>	43.2±7.5
	PFYL	0.177±0.047	59.9±3.2	0.946±0.274	47.6±12.0	2.435±0.847	74.8±22.9	5.236±2.331	49.1±12.1
	DBB	0.509±0.109	116.4±34.1	2.131±0.597	139.3±39.1	7.003±1.936	98.7±34.3	14.828±3.086	85.2±17.2
	LAVA	0.104±0.036	44.8±8.1	0.851±0.385	40.9±3.9	2.483±0.997	38.1±2.4	6.189±2.365	44.5±6.0
	PEAR (Ours)	0.104±0.038	<u>28.4±5.4</u>	<u>0.774±0.268</u>	<u>23.5±0.4</u>	2.138±0.762	<u>35.1±4.9</u>	4.246±1.106	<u>38.4±2.5</u>
Knapsack	MSE	0.351±0.020	422.8±44.5	0.922±0.066	180.8±6.2	1.717±0.226	111.7±11.8	2.285±0.205	99.1±15.8
	SPO+	<u>0.291±0.015</u>	600.0±0.0	<u>0.381±0.036</u>	600.0±0.0	<u>0.608±0.066</u>	600.0±0.0	<u>0.763±0.075</u>	600.0±0.0
	PFYL	0.506±0.025	600.0±0.0	0.442±0.042	600.0±0.0	0.723±0.049	600.0±0.0	1.056±0.147	600.0±0.0
	DBB	1.852±0.280	600.0±0.0	0.910±0.160	600.0±0.0	1.040±0.307	600.0±0.0	1.081±0.128	600.0±0.0
	LAVA	0.438±0.045	600.0±0.0	0.478±0.021	516.9±81.1	0.820±0.060	332.5±111.0	1.351±0.223	<u>182.8±42.5</u>
	PEAR (Ours)	0.287±0.007	<u>567.9±45.1</u>	0.315±0.020	<u>305.2±40.8</u>	0.388±0.032	<u>261.7±37.4</u>	0.437±0.043	271.7±57.0

Lee, J., Jin, S., & Lee, Y. (2026). Decision-Focused Learning via Tangent-Space Projection of Prediction Error. *Forty-Third International Conference on Machine Learning (ICML 2026)*

Decision-focused learning

- If the prediction model is overly optimized for decision quality, its prediction quality can be critically degraded
 - may be useless for the other downstream tasks

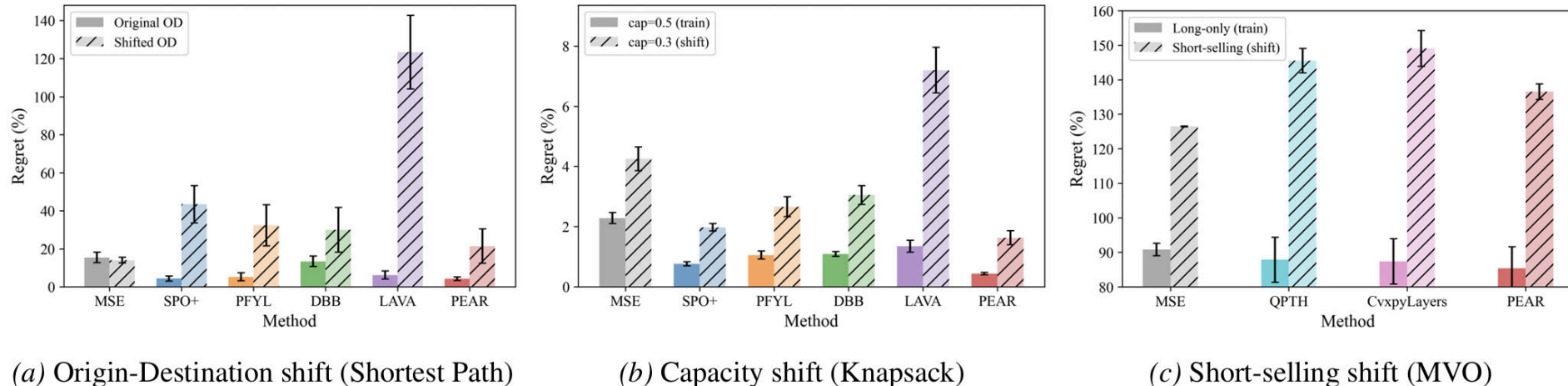


Figure 2. **Robustness under constraint shifts.** We train on an original train set and evaluate under shifted constraints. Our method maintains the lowest regret among DFL methods across all three tasks. Additional results are in Appendix F.

Decision-focused learning

- If the prediction model is overly optimized for decision quality, its prediction quality can be critically degraded
 - may be useless for the other downstream tasks
 - often suggested to add a regularization term in decision loss minimization

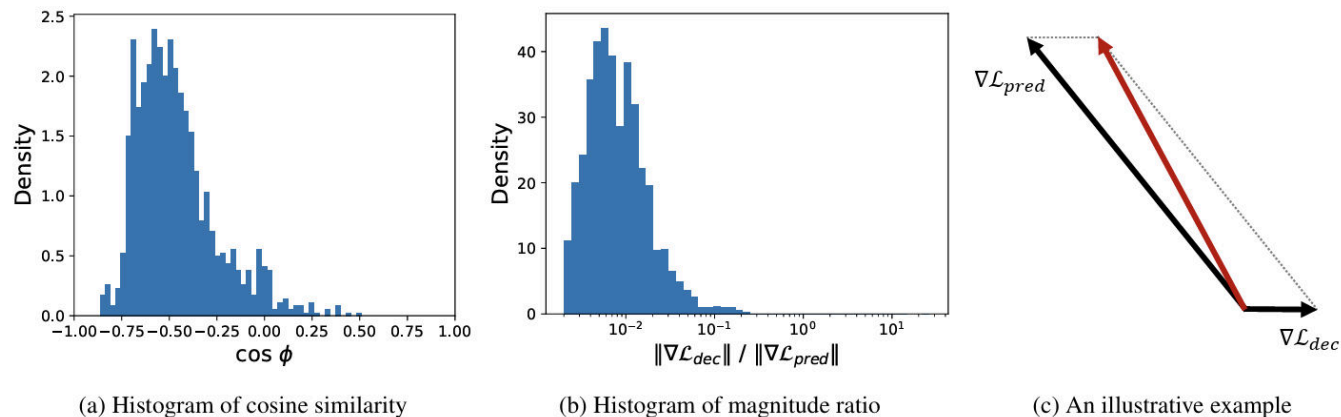


Figure 2: Comparison of prediction loss gradients $\nabla\mathcal{L}_{pred}$ and decision loss gradients $\nabla\mathcal{L}_{dec}$ during training on the budget allocation problem. Each subplot represents: (a) the cosine similarity between $\nabla\mathcal{L}_{pred}$ and $\nabla\mathcal{L}_{dec}$, (b) the gradient norm ratio $\|\nabla\mathcal{L}_{dec}\|/\|\nabla\mathcal{L}_{pred}\|$ on a log scale, and (c) an illustrative case where two gradients with characteristics of (a) and (b) are naively added. In subplot (a), cosine similarity values near 1 indicate that the gradients are aligned in the same direction, while values near -1 indicate they point in opposite directions. The figure shows that in most cases, $\cos \phi < 0$ for budget allocation, meaning the gradients are often conflicting. Subplot (b) shows that in many instances, the magnitude of $\nabla\mathcal{L}_{dec}$ is approximately 100 times smaller than that of $\nabla\mathcal{L}_{pred}$. Subplot (c) demonstrates that naively summing two conflicting and imbalanced may lead to biased training.



Return Prediction for Mean-Variance Portfolio Selection: How Decision-Focused Learning Shapes Forecasting Models

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Abstract

Markowitz laid the foundation of portfolio theory through mean-variance optimization (MVO). However, MVO's effectiveness depends on precise estimation of expected returns, variances, and covariances, which are typically uncertain. Machine learning models are increasingly used to estimate these parameters, trained to minimize prediction errors like MSE, which treats errors uniformly across assets. Recent studies show this leads to suboptimal decisions and propose Decision-Focused Learning (DFL), integrating prediction and optimization to improve outcomes. While studies demonstrate DFL's potential to enhance portfolio performance, the mechanisms of how DFL modifies prediction models for MVO remain unexplored. This study investigates how DFL adjusts stock return prediction models to optimize MVO decisions. We show that DFL's gradient tilts MSE-based prediction errors by the inverse covariance matrix Σ^{-1} , incorporating inter-asset correlations into learning, while MSE treats each asset independently. This tilting creates systematic biases where DFL overestimates returns for portfolio assets while underestimating excluded assets. Our findings reveal why DFL achieves superior performance despite higher prediction errors. The strategic biases are features, not flaws.

1 Introduction

Decision-making is a crucial aspect across various fields, where optimization is often employed to guide us toward the best possible outcome quantitatively. In ideal scenarios, where the values of all parameters in the optimization formula are known, a mathematically optimal solution can be determined with precision. However, in most real-world situations, parameters are often uncertain. The quality of the decisions derived from optimization heavily depends on how accurately these uncertain parameters are estimated. Therefore, it is essential to ensure accurate estimation of these parameters to achieve high-quality decision-making outcomes.

This is particularly evident in the field of portfolio optimization, where financial decisions are made under uncertainty. Harry Markowitz laid the foundation of portfolio theory based on mean-variance optimization (MVO) [24]. The objective of MVO is to construct an investment portfolio that maximizes return for a given level of risk or minimizes risk for a given level of expected return. While the MVO has been fundamental to investment management [19], its effectiveness depends on the accurate estimates of the expected return, variance, and covariance of asset returns, which are often uncertain in practice. Despite this uncertainty, decision makers who utilize MVO should have their own estimates of expected

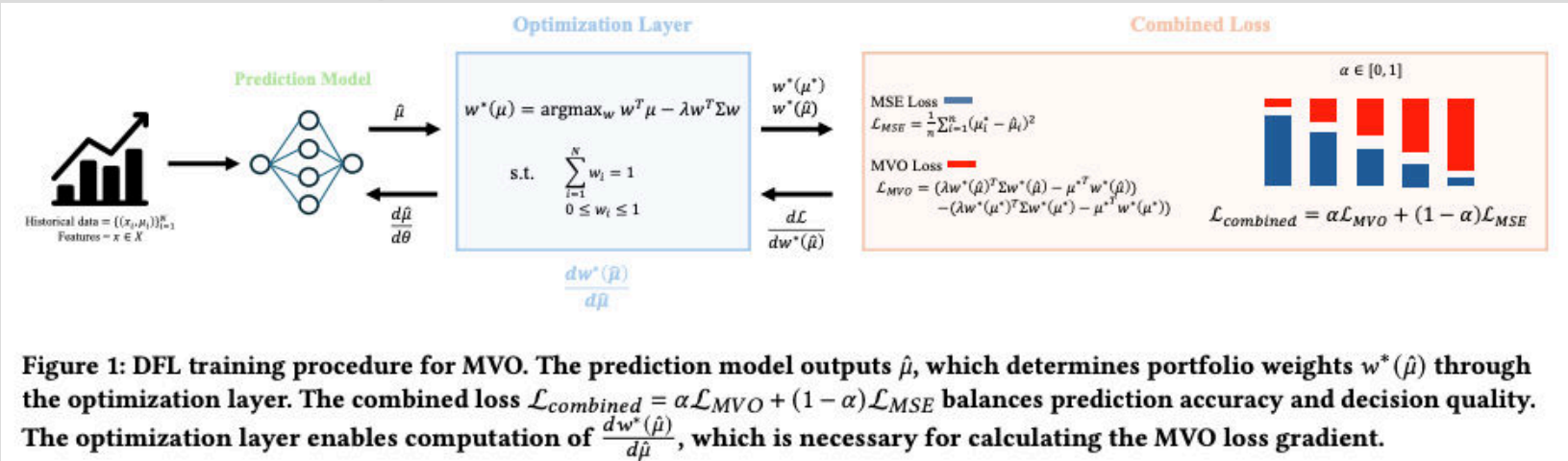
Paper 1



Return Prediction for Mean-Variance Portfolio Selection: How Decision-Focused Learning Shapes Forecasting Models

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to demonstrate DFL's potential to enhance portfolio performance, the mechanisms of how DFL modifies prediction models for MVO remain unexplored. This study investigates how DFL adjusts stock return prediction models to optimize MVO decisions. We show that DFL's gradient tilts MSE-based prediction errors by the inverse covariance matrix Σ^{-1} , incorporating inter-asset correlations into learning, while MSE treats each asset independently. This tilting creates systematic biases where DFL overestimates returns for portfolio assets while underestimating excluded assets. Our findings reveal why DFL achieves superior performance despite higher prediction errors. The strategic biases are features, not flaws.

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Paper 1



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$$\mathcal{L}_{\text{Combined}} = \alpha \mathcal{L}_{\text{MVO}} + (1 - \alpha) \mathcal{L}_{\text{MSE}} \quad (12)$$

Table 2: Portfolio performance metrics for varying λ and α . Models with $\alpha > 0$ consistently outperform pure MSE ($\alpha = 0$), with optimal performance typically at intermediate values ($\alpha \in [0.25, 0.75]$). DFL improves both returns and risk metrics, achieving higher Sharpe ratios and often lower maximum drawdowns. Bold values indicate best performance for each metrics.

Panel A. DOW 30 Dataset						Panel B. S&P 100 Dataset					
λ	α	Return ($\uparrow$)	Sharpe ($\uparrow$)	MDD ($\downarrow$)	Wealth ($\uparrow$)	λ	α	Return ($\uparrow$)	Sharpe ($\uparrow$)	MDD ($\downarrow$)	Wealth ($\uparrow$)
0.1	0.00	0.181	0.681	0.253	1.459	0.1	0.00	0.209	0.734	0.266	1.184
	0.25	0.323	1.138	0.230	1.669		0.25	0.256	0.744	0.327	1.673
	0.50	0.398	1.228	0.261	2.138		0.50	0.134	0.513	0.278	1.239
	0.75	0.177	0.699	0.268	1.408		0.75	0.091	0.386	0.319	1.277
	1.00	0.143	0.593	0.226	1.275		1.00	0.292	1.179	0.249	1.701
0.5	0.00	0.125	0.529	0.251	1.250	0.5	0.00	0.153	0.578	0.276	1.126
	0.25	0.192	0.754	0.258	1.459		0.25	0.187	0.769	0.295	1.339
	0.50	0.347	1.302	0.211	1.813		0.50	0.231	0.701	0.295	1.680
	0.75	0.248	1.051	0.230	1.520		0.75	0.303	1.217	0.230	1.763
	1.00	0.151	0.641	0.268	1.354		1.00	0.183	0.769	0.267	1.512
1.0	0.00	0.115	0.519	0.242	1.234	1.0	0.00	0.129	0.540	0.242	1.090
	0.25	0.165	0.775	0.221	1.383		0.25	0.237	0.746	0.267	1.772
	0.50	0.327	1.150	0.218	1.669		0.50	0.319	1.260	0.214	1.773
	0.75	0.312	1.311	0.203	1.651		0.75	0.201	0.801	0.237	1.410
	1.00	0.180	0.763	0.256	1.397		1.00	0.252	0.949	0.287	1.569
5.0	0.00	0.091	0.531	0.195	1.163	5.0	0.00	0.144	0.838	0.194	1.258
	0.25	0.217	0.932	0.203	1.566		0.25	0.203	0.831	0.250	1.523
	0.50	0.187	0.756	0.215	1.473		0.50	0.136	0.585	0.219	1.347
	0.75	0.197	0.871	0.211	1.379		0.75	0.193	0.855	0.237	1.464
	1.00	0.164	0.738	0.223	1.361		1.00	0.213	0.928	0.276	1.611

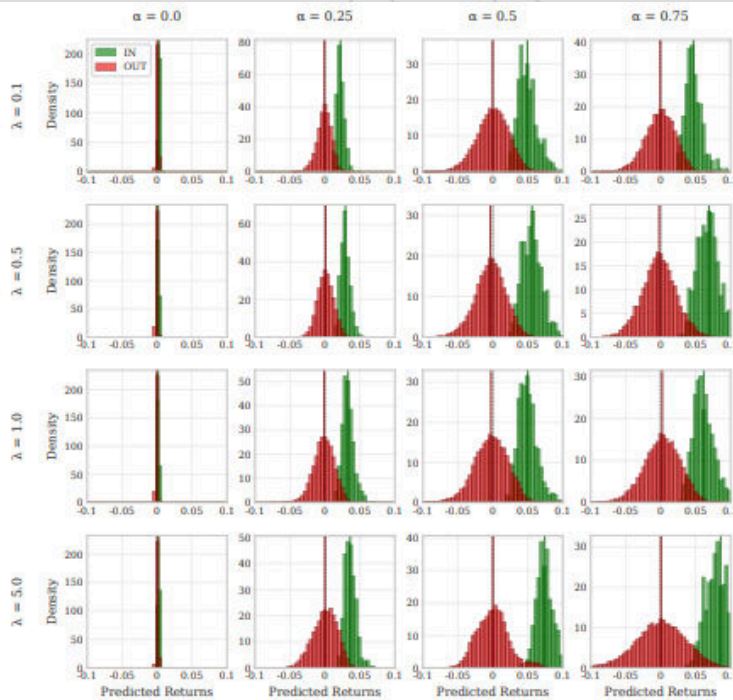
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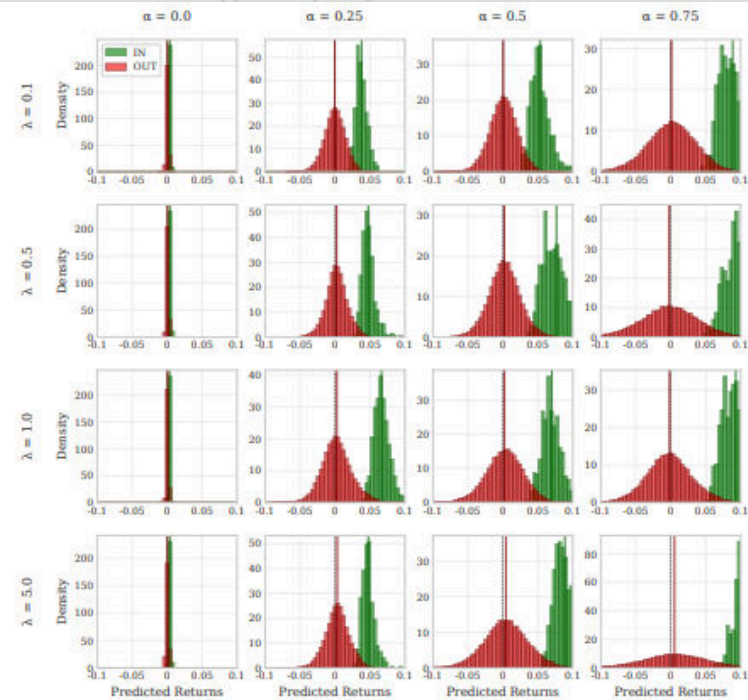
Paper 1



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(a) DOW30



(b) S&P100

Figure 4: Predicted return distributions for IN/OUT portfolio groups across different λ and α values. As α increases, the separation between IN and OUT group distributions widens. The case of $\alpha = 1$ is excluded due to extreme distribution separation.

ings reveal why DFL achieves superior performance despite higher prediction errors. The strategic biases are features, not flaws.

often uncertain in practice. Despite this uncertainty, decision makers who utilize MVO should have their own estimates of expected



Estimating Covariance for Global Minimum Variance Portfolio: A Decision-Focused Learning Approach

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Abstract

Portfolio optimization constitutes a cornerstone of risk management by quantifying the risk-return trade-off. Since it inherently depends on accurate parameter estimation under conditions of future uncertainty, the selection of appropriate input parameters is critical for effective portfolio construction. However, most conventional statistical estimators and machine learning algorithms determine these parameters by minimizing mean-squared error (MSE), a criterion that can yield suboptimal investment decisions.

In this paper, we adopt decision-focused learning (DFL)—an approach that directly optimizes decision quality rather than prediction error such as MSE—to derive the global minimum-variance portfolio (GMVP). Specifically, we theoretically derive the gradient of decision loss using the analytic solution of GMVP and its properties regarding the principal components of itself. Through extensive empirical evaluation, we show that prediction-focused estimation methods may fail to produce optimal allocations in practice, whereas DFL-based methods consistently deliver superior decision performance. Furthermore, we provide a comprehensive analysis of DFL’s mechanism in GMVP construction, focusing on its volatility reduction capability, decision-driving features, and estimation characteristics.

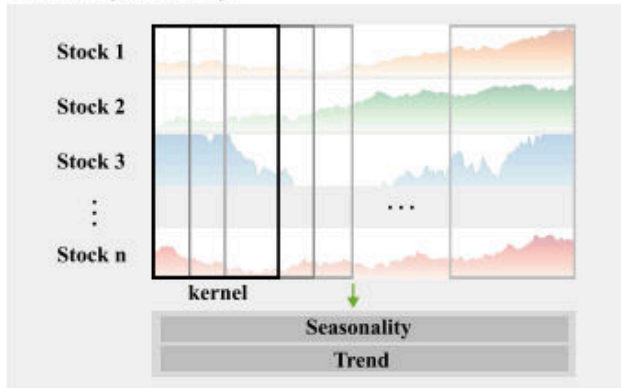
1 Introduction

Markowitz’s seminal work introduced mean-variance optimization (MVO) [29], which provides a quantitative framework for balancing risk and return and has since become modern portfolio theory’s foundation. The most important part in MVO is parameter estimation, due to its high sensitivity [21]. Practically, MVO parameters are typically calibrated using an in-sample period under the assumption of return stationarity, which results in poor performances because the distribution underlying the in-sample and out-of-sample periods often differs. Empirical studies have shown that MVO portfolios constructed with estimated parameters can underperform naïve diversification strategies like the equally weighted portfolio, highlighting appropriate parameter estimation [15].

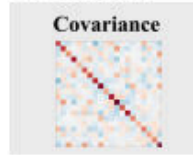
Global minimum-variance portfolio (GMVP) constitutes the least risky portfolio in the efficient frontier, therefore it inherits a key for understanding the modern portfolio theory. Since GMVP depends solely on the covariance matrix, not on expected returns, its parameter uncertainty becomes much lower than in the MVO framework. Moreover, empirical evidence indicates that GMVPs often achieve superior out-of-sample return performance compared to market-cap-weighted benchmarks [12, 13] and exhibit lower realized volatility [6]. Because GMVP construction requires estimating only covariances—unlike MVO, which requires both ex-

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Model (DLinear)



Prediction



$$\frac{d\hat{\Sigma}}{d\theta} \quad \frac{dw^*(\hat{\Sigma})}{d\hat{\Sigma}} \quad \frac{d\mathcal{L}}{dw^*(\hat{\Sigma})}$$

Forward Backpropagation

Solve GMVP

Optimization

$$w^*(\Sigma) = \underset{w}{\operatorname{argmin}} w^\top \Sigma w$$

$$\text{s.t. } \sum_{i=1}^n w_i = 1$$

Analytic solution

$$w^*(\Sigma) = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}$$

$$w^*(\hat{\Sigma}) \downarrow \quad \downarrow w^*(\Sigma_{\text{true}})$$

Loss

$$w^*(\hat{\Sigma})^\top \Sigma_{\text{true}} w^*(\hat{\Sigma})$$

$$-w^*(\Sigma_{\text{true}})^\top \Sigma_{\text{true}} w^*(\Sigma_{\text{true}})$$

- Estimated - optimal



Figure 1: DFL framework for GMVP construction. Historical returns of N assets are processed by DLinear to predict covariance matrix Σ . The GMVP optimization layer computes portfolio weights w^* using the closed form solution, and the regret loss is applied to minimize the gap between predicted and optimal portfolio volatility for end-to-end training.

properties regarding the principal components of itself. Through extensive empirical evaluation, we show that prediction-focused estimation methods may fail to produce optimal allocations in practice, whereas DFL-based methods consistently deliver superior decision performance. Furthermore, we provide a comprehensive analysis of DFL's mechanism in GMVP construction, focusing on its volatility reduction capability, decision-driving features, and estimation characteristics.

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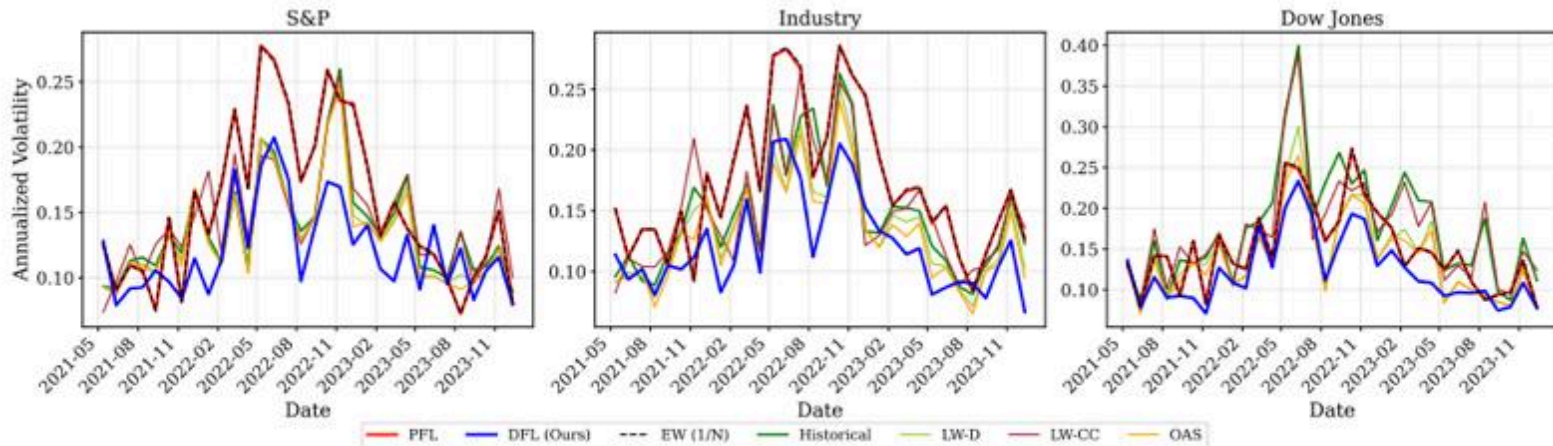


Figure 3: Annualized volatility of each model's portfolio with $\delta_{in} = \delta_{out} = 21$ from the left to the right, respectively.

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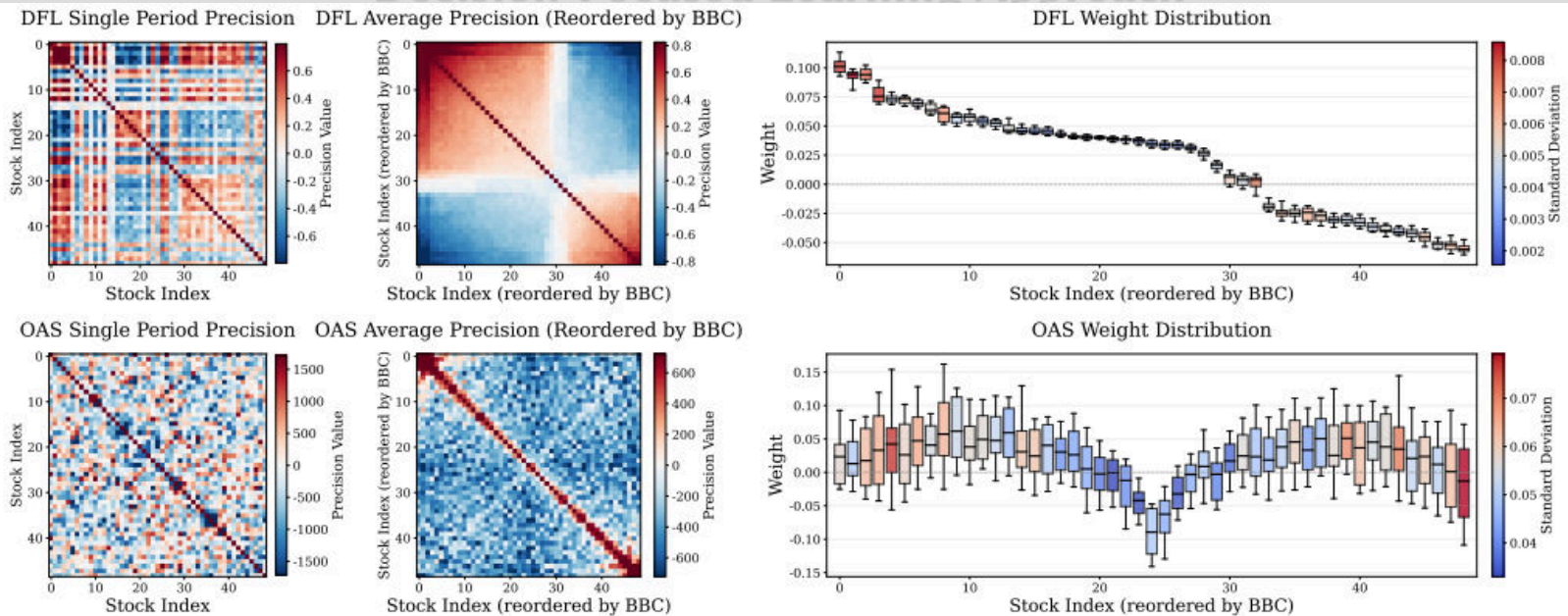


Figure 6: Precision matrix derived by DFL and OAS for single test period in the Industry dataset (left), Average of the permuted precision matrix estimated by DFL and OAS for the entire test period in the Industry dataset (middle), Average and standard deviation of GMVP weights in the reordered index for the entire test period in the Industry dataset (right).

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